

GEMM Analytical Model - Equations

Auto-generated from PYTHON_FORMULAS and LATEX_SYMBOL_OVERRIDES in gemmanalytics.py.

Output order: execution

Machine Parameters (Pre-Defined)

- [row 8] $\$_{GHz}^{(GT)}$: GT Freq (GHz)
- [row 26] $\$$: XeCU count
- [row 27] $\$_{perXeCU}$: XeCore per XeCU
- [row 28] $\$_{perXeCore}$: EU per XeCore
- [row 29] $\$_{perXeCU}$: L2 banks per XeCU
- [row 30] $\$_{MB}$: bank capacity (MB)
- [row 32] $\$_{DPAS}$: DPAS depth
- [row 35] $\$_{systolic}$: compute efficiency %
- [row 86] $\$_{Bp(Clk)EU}^{max}$: L1 read max (B/EU/clock)
- [row 87] $\$_{Bp(Clk)EU}^{max}$: L1 write max (B/EU/clock)
- [row 97] $\$_{BpClk}^{max}$: GTI read max BW (B/clock)
- [row 98] $\$_{BpClk}^{max}$: GTI write max BW (B/clock)
- [row 124] $\$_{GBps}^{max}$: max possible HBM BW (GB/s)

Workload Parameters (Pre-Defined)

- [row 3] $\$_{matA}$: input A data format (e.g. fp8)
- [row 4] $\$_{matB}$: input B data format (e.g. fp4)
- [row 5] $\$_{matD}$: output D data format (e.g. fp8)
- [row 9] $\$_{dim}$: M dimension
- [row 10] $\$_{dim}$: K dimension
- [row 11] $\$_{dim}$: N dimension
- [row 24] $\$_{perElement}^{matD}$: output bytes per element (fp32)
- [row 34] $\$_{GT}$: machine occupancy %
- [row 39] $\$_{perThread}$: M per thread
- [row 40] $\$_{perThread}$: K per thread
- [row 41] $\$_{perThread}$: N per thread
- [row 45] $\$_{thread}^{(ThreadGroup)}$: TG width in units of thread

[row 46] \$ _{thread}^{\{(ThreadGroup)\}} \$: TG height in units of thread

[row 50] \$ _{TG,keep cluster size as 4}^{\{(XeCoreCluster)\}} \$: XeCore cluster width in units of TG (keep cluster size as 4)

[row 51] \$ _{TG,keep cluster size as 4}^{\{(XeCoreCluster)\}} \$: XeCore cluster height in units of TG (keep cluster size as 4)

[row 54] \$ _{TG}^{\{(XECUTile)\}} \$: XeCU tile width in units of TG

[row 55] \$ _{TG}^{\{(XECUTile)\}} \$: XeCU tile height in units of TG

[row 58] \$ _{XeCU}^{\{(GPUPTile)\}} \$: GPU tile width in XeCU unit

Computed Equations

row 22 - INPUT A BYTES PER ELEMENT

$$\text{Bytes}_{perElement}^{(matA)} = \text{DataFormatToBytes}[\text{Fmt}^{(matA)}]$$

row 23 - INPUT B BYTES PER ELEMENT

$$\text{Bytes}_{perElement}^{(matB)} = \text{DataFormatToBytes}[\text{Fmt}^{(matB)}]$$

row 25 - OUTPUT BYTES PER ELEMENT AFTER DOWN CONVERSION

$$\text{Bytes}_{perElement}^{(matD\downarrow)} = \text{DataFormatToBytes}[\text{Fmt}^{(matD\downarrow)}]$$

row 33 - EU COUNT

$$|\text{EU}| = |\text{XeCore}|_{perXeCU} \times |\text{EU}|_{perXeCore} \times |\text{XeCU}|$$

row 36a - MMA MAC THROUGHPUT PER EU

$$\tau_{mMACp(Cl k \cdot EU)}^{(peak)} = \frac{4}{\max(\text{Bytes}_{perElement}^{(matA)}, \text{Bytes}_{perElement}^{(matB)})} \times D_{DPAS} \times 16$$

row 36 - MMA MAC THROUGHPUT PER XECORE

$$\tau_{mMACp(Cl k \cdot XeCore)}^{(peak)} = \tau_{mMACp(Cl k \cdot EU)}^{(peak)} \times |\text{EU}|_{perXeCore}$$

row 42 - CLKS PER DPAS

$$\text{CLKS}_{DPAS} = \frac{M_{perThread} \times K_{perThread} \times N_{perThread}}{\frac{\tau_{mMACp(Cl k \cdot XeCore)}^{(peak)}}{|\text{EU}|_{perXeCore}}}$$

row 43 - THREAD WIDTH IN UNITS OF ELEMENTS

$$W_{element}^{(Thread)} = N_{perThread}$$

row 44 - THREAD HEIGHT IN UNITS OF ELEMENTS

$$H_{element}^{(Thread)} = M_{perThread}$$

row 47 - TG WIDTH IN UNITS OF ELEMENT REALIZED BY MULTIPLE MMA ITERATIONS

$$W_{element}^{(ThreadGroup, realized by multiple MMA iterations)} = W_{thread}^{(ThreadGroup)} \times W_{element}^{(Thread)}$$

row 14 - TG TILES IN N

$$|Tiles|_N^{(TG)} = \frac{N_{dim}}{W_{element}^{(ThreadGroup, realized by multiple MMA iterations)}}$$

row 48 - TG HEIGHT IN UNITS OF ELEMENT

$$H_{element}^{(ThreadGroup)} = H_{thread}^{(ThreadGroup)} \times H_{element}^{(Thread)}$$

row 15 - TG TILES IN M

$$|Tiles|_M^{(TG)} = \frac{M_{dim}}{H_{element}^{(ThreadGroup)}}$$

row 52 - XECORE CLUSTER WIDTH IN UNITS OF ELEMENT

$$W_{element}^{(XeCoreCluster)} = W_{TG, keep cluster size as 4}^{(XeCoreCluster)} \times W_{element}^{(ThreadGroup, realized by multiple MMA iterations)}$$

row 16 - TG CLUSTER TILES IN N

$$|Tiles|_N^{(TG Cluster)} = \frac{N_{dim}}{W_{element}^{(XeCoreCluster)}}$$

row 53 - XECORE CLUSTER HEIGHT IN UNITS OF ELEMENT

$$H_{element}^{(XeCoreCluster)} = H_{TG, keep cluster size as 4}^{(XeCoreCluster)} \times H_{element}^{(ThreadGroup)}$$

row 17 - TG CLUSTER TILES IN M

$$|Tiles|_M^{(TG Cluster)} = \frac{M_{dim}}{H_{element}^{(XeCoreCluster)}}$$

row 56 - XECU TILE WIDTH IN UNITS OF ELEMENT

$$W_{element}^{(XeCUTile)} = W_{TG}^{(XECUTile)} \times W_{element}^{(ThreadGroup, realized by multiple MMA iterations)}$$

row 18 - XECU TILES IN N

$$|Tiles|_N^{(XeCU)} = \frac{N_{dim}}{W_{element}^{(XeCUTile)}}$$

row 20 - GPU TILES IN N

$$|\text{Tiles}|_N^{(GPU)} = \frac{N_{dim}}{W_{element}^{(XeCUTile)}} \times W_{XeCU}^{(GPUTile)}$$

row 57 - XECU TILE HEIGHT IN UNITS OF ELEMENT

$$H_{element}^{(XeCUTile)} = H_{TG}^{(XECUTile)} \times H_{element}^{(ThreadGroup)}$$

row 19 - XECU TILES IN M

$$|\text{Tiles}|_M^{(XeCU)} = \frac{M_{dim}}{H_{element}^{(XeCUTile)}}$$

row 59 - GPU TILE HEIGHT IN XECU UINT

$$H_{XeCU}^{(GPUTile)} = \frac{|\text{XeCU}|}{W_{XeCU}^{(GPUTile)}}$$

row 21 - GPU TILES IN M

$$|\text{Tiles}|_M^{(GPU)} = \frac{M_{dim}}{H_{element}^{(XeCUTile)}} \times H_{XeCU}^{(GPUTile)}$$

row 13 - WAVES

$$N_{waves} = \text{CEILING}(|\text{Tiles}|_N^{(GPU)}, 1) \times \text{CEILING}(|\text{Tiles}|_M^{(GPU)}, 1)$$

row 60 - GPU TILE WIDTH IN UNITS OF ELEMETNS

$$W_{element}^{(GPUTile)} = W_{XeCU}^{(GPUTile)} \times W_{element}^{(XeCUTile)}$$

row 61 - GPU TILE HEIGHT IN UNITS OF ELEMETNS

$$H_{element}^{(GPUTile)} = H_{XeCU}^{(GPUTile)} \times H_{element}^{(XeCUTile)}$$

row 63 - MAT A INPUT SIZE B

$$\text{Size}_B^{(matA)} = M_{dim} \times K_{dim} \times \text{Bytes}_{perElement}^{(matA)}$$

row 64 - MAT B INPUT SIZE B

$$\text{Size}_B^{(matB)} = K_{dim} \times N_{dim} \times \text{Bytes}_{perElement}^{(matB)}$$

row 65 - MAT C INPUT D OUTPUT SIZE B

$$\text{Size}_B^{(matC, matD)} = M_{dim} \times N_{dim} \times \text{Bytes}_{perElement}^{(matD \downarrow)}$$

row 66 - MAT D INTERMEDIATE SIZE B

$$\text{Size}_B^{(matD\downarrow)} = M_{dim} \times N_{dim} \times \text{Byte}_{perElement}^{(matD)}$$

row 102 - TOTAL L2 SIZE B FOR A SINGLE INSTANCE

$$\text{L2Size}_B = \text{L2BankSize}_{MB} \times |\text{L2Banks}|_{perXeCU} \times 1024 \times 1024$$

**row 103 - WORKING DATA SET SIZE OF K IN L2 CORRESP 20K CLOCKS OF
THREAD DIVERGENCE**

$$|\text{WorkingSet}|_{20K\text{ clks of thread divergence}}^{(K\text{ in }L2)} = \min(20000 \times \frac{K_{perThread}}{\text{CLKS}_{DPAS}}, K_{dim})$$

**row 104 - TOTAL REQUIRED L2 SIZE FOR IDEAL HIT RATE B FOR A SINGLE
INSTANCE AND SINGLE WAVE**

$$\begin{aligned} \text{TotalRequiredL2Size}_{matB\text{ always hit}}^{1\text{ instance, 1 wave}} &= H_{element}^{(XeCUTile)} \times \text{Bytes}_{perElement}^{(matA)} \times |\text{WorkingSet}|_{20K\text{ clks of thread divergence}}^{(K\text{ in }L2)} \\ &+ W_{element}^{(XeCUTile)} \times \text{Bytes}_{perElement}^{(matB)} \times |\text{WorkingSet}|_{20K\text{ clks of thread divergence}}^{(K\text{ in }L2)} \\ &+ W_{element}^{(XeCUTile)} \times H_{element}^{(XeCUTile)} \times \text{Bytes}_{perElement}^{(matD\downarrow)} \end{aligned}$$

row 125 - MAX POSSIBLE HBM BW FREQ B CLK

$$\text{MemBW}_{BpClk}^{max} = \frac{\text{MemBW}_{GBps}^{max}}{f_{GHz}^{(GT)}}$$

row 37a - WORKLOAD MAC PER XECORE

$$\text{WL}_{MAC}^{(XeCore)} = \frac{M_{dim} \times K_{dim} \times N_{dim}}{|\text{XeCore}|_{perXeCU} \times |\text{XeCU}|}$$

row 37 - CLK SPECIFIED EFFICIENCY

$$T_{clk}^{(total)} = \frac{\text{WL}_{MAC}^{(XeCore)}}{\tau_{mMACp(Cl k \cdot XeCore)}^{(peak)} \times \eta_{systolic}}$$

row 69 - TOTAL L2 READ B

$$\begin{aligned} \text{L2Rd}_B^{(total)} &= \text{Size}_B^{(matA)} \times \text{CEILING}(\frac{N_{dim}}{W_{element}^{(ThreadGroup, realized by multiple MMA iterations)}}, 1) \\ &+ \text{Size}_B^{(matB)} \times \text{CEILING}(\frac{M_{dim}}{H_{element}^{(ThreadGroup)}}), 1) \end{aligned}$$

row 70 - TOTAL L2 WRITE B

$$\text{L2Wr}_B^{(total)} = \text{Size}_B^{(matC, matD)}$$

row 71 - TOTAL L1 READ B

$$\begin{aligned} \text{L1Rd}_B^{(total)} = & \text{Size}_B^{(matA)} \times \text{CEILING}\left(\frac{N_{dim}}{W_{element}^{(Thread)}}, 1\right) \\ & + \text{Size}_B^{(matB)} \times \text{CEILING}\left(\frac{M_{dim}}{H_{element}^{(Thread)}}, 1\right) \end{aligned}$$

row 72 - TOTAL L1 WRITE B

$$\text{L1Wr}_B^{(total)} = \text{Size}_B^{(matC, matD)}$$

row 74 - L2 READ B XECORE CLK

$$\text{L2RdBW}_{Bp(Clk \cdot XeCore)} = \frac{\frac{\text{L2Rd}_B^{(total)}}{|\text{XeCU}| \times |\text{XeCore}|_{perXeCU}}}{T_{clk}^{(total)}}$$

row 75 - L2 WRITE B XECORE CLK

$$\text{L2WrBW}_{Bp(Clk \cdot XeCore)} = \frac{\frac{\text{L2Wr}_B^{(total)}}{|\text{XeCU}| \times |\text{XeCore}|_{perXeCU}}}{T_{clk}^{(total)}}$$

row 76 - L2 READ WRITE B XECORE CLK

$$\begin{aligned} \text{L2RdWrBW}_{Bp(Clk \cdot XeCore)} = & \text{L2RdBW}_{Bp(Clk \cdot XeCore)} \\ & + \text{L2WrBW}_{Bp(Clk \cdot XeCore)} \end{aligned}$$

row 79 - L1 READ B EU CLK

$$\text{L1RdBW}_{Bp(Clk \cdot EU)} = \frac{\frac{\text{L1Rd}_B^{(total)}}{|\text{EU}|}}{T_{clk}^{(total)}}$$

row 77 - L1 READ B XECORE CLK

$$\text{L1RdBW}_{Bp(Clk \cdot XeCore)} = \text{L1RdBW}_{Bp(Clk \cdot EU)} \times |\text{EU}|_{perXeCore}$$

row 80 - L1 WRITE B EU CLK

$$\text{L1WrBW}_{Bp(Clk \cdot EU)} = \frac{\frac{\text{L1Wr}_B^{(total)}}{|\text{EU}|}}{T_{clk}^{(total)}}$$

row 78 - L1 WRITE B XECORECLK

$$\text{L1WrBW}_{Bp(Clk \cdot XeCore)} = \text{L1WrBW}_{Bp(Clk \cdot EU)} \times |\text{EU}|_{perXeCore}$$

row 83 - L2 READ MAX B XECORE CLK

$$L2RdBW_{Bp(Clk \cdot XeCore)}^{(max)} = \frac{|XeCore|_{perXeCU} \times 64}{|XeCore|_{perXeCU}}$$

row 84 - L2 WRITE MAX B XECORE CLK

$$L2WrBW_{Bp(Clk \cdot XeCore)}^{(max)} = L2RdBW_{Bp(Clk \cdot XeCore)}^{(max)}$$

row 85 - L2 READ WRITE MAX B XECORE CLK

$$L2RdWrBW_{Bp(Clk \cdot XeCore)}^{(max)} = L2WrBW_{Bp(Clk \cdot XeCore)}^{(max)}$$

row 90 - L2 READ B XECORE CLK PCT

$$\eta_{L2RdBWpXeCore} = \frac{L2RdBW_{Bp(Clk \cdot XeCore)}}{L2RdBW_{Bp(Clk \cdot XeCore)}^{(max)}}$$

row 91 - L2 WRITE B XECORE CLK PCT

$$\eta_{L2WrBWpXeCore} = \frac{L2WrBW_{Bp(Clk \cdot XeCore)}}{L2WrBW_{Bp(Clk \cdot XeCore)}^{(max)}}$$

row 92 - L2 READ WRITE B XECORE CLK PCT

$$\eta_{L2RdWrBWpXeCore} = \frac{L2RdWrBW_{Bp(Clk \cdot XeCore)}}{L2RdWrBW_{Bp(Clk \cdot XeCore)}^{(max)}}$$

row 93 - L1 READ B EU CLK PCT

$$\eta_{L1RdBWpEU} = \frac{L1RdBW_{Bp(Clk \cdot EU)}}{L1RdBW_{Bp(Clk \cdot EU)}^{max}}$$

row 94 - L1 WRITE B EU CLK PCT

$$\eta_{L1WrBWpEU} = \frac{L1WrBW_{Bp(Clk \cdot EU)}}{L1WrBW_{Bp(Clk \cdot EU)}^{max}}$$

row 106 - L2 HIT RATE ASSUMED RANDOM ACCESS WITHIN THE WORKING DATA SET PCT

$$L2HitRate_{random\ WS\ access} = \min\left(\frac{L2Size_B}{TotalRequiredL2Size_{1\ instance,\ 1\ wave\ mat\ B\ always\ hit}}, 1\right)$$

row 107 - L2 MISS RATE PCT

$$L2MissRate = 1 - L2HitRate_{random\ WS\ access}$$

row 108 - TOTAL L2 READ TRAFFIC B

$$L2RdB_{total} = L2RdB^{(total)}$$

row 110 - PROBABILITY OF MATA HIT IN L2 DURING A NON FIRST WAVE

PCT

$$\begin{aligned} P_{L2Hit, after\ 1st\ wave}^{(matA)} = & IF(Size_B^{(matA)} \\ & + Size_B^{(matB)} \\ & + Size_B^{(matC, matD)} \leq L2Size_B, 1.0, IF(K_{dim} > 2 \times |WorkingSet|_{20K\ clks\ of\ thread\ divergence}^{(K\ in\ L2)}, 0.0, 1 \\ & - \frac{K_{dim} - |WorkingSet|_{20K\ clks\ of\ thread\ divergence}^{(K\ in\ L2)}}{|WorkingSet|_{20K\ clks\ of\ thread\ divergence}^{(K\ in\ L2)}})) \end{aligned}$$

row 111 - PROBABILITY OF MATB HIT IN L2 DURING A NON FIRST WAVE

PCT

$$\begin{aligned} P_{L2Hit, after\ 1st\ wave}^{(matB)} = & IF(Size_B^{(matA)} \\ & + Size_B^{(matB)} \\ & + Size_B^{(matC, matD)} \leq L2Size_B, 1.0, 0.0) \end{aligned}$$

row 112 - PROBABILITY OF MATA MISS IN L2 DURING A NON FIRST WAVE

PCT

$$\begin{aligned} P_{L2Miss, after\ 1st\ wave}^{(matA)} = & 1 \\ & - P_{L2Hit, after\ 1st\ wave}^{(matA)} \end{aligned}$$

row 113 - PROBABILITY OF MATB MISS IN L2 DURING A NON FIRST WAVE

PCT

$$\begin{aligned} P_{L2Miss, after\ 1st\ wave}^{(matB)} = & 1 \\ & - P_{L2Hit, after\ 1st\ wave}^{(matB)} \end{aligned}$$

row 116 - TOTAL HBM READ B AFTER A COMPLETION OF A WAVE CONSIDER

COLD CACHE

$$MemRdB_{cold\ start}^{(wave)} = \frac{\left(MemRdB_{cold\ start}^{(wave)} \right)_{num}}{2}$$

$$\begin{aligned}
\left(\text{MemRdB}_{cold\ start}^{(wave)}\right)_{num} &= \text{Size}_B^{(matA)} \\
&+ \text{Size}_B^{(matA)} \times \left(\text{CEILING}\left(\frac{N_{dim}}{W_{(XeCUTile)}}, 1\right) - 1\right) \times P_{L2Miss, after\ 1st\ wave}^{(matA)} \\
&+ \text{Size}_B^{(matB)} \\
&+ \text{Size}_B^{(matB)} \times \left(\text{CEILING}\left(\frac{M_{dim}}{H_{(XeCUTile)}}, 1\right) - 1\right) \times P_{L2Miss, after\ 1st\ wave}^{(matB)} \\
&+ \left(\left(\text{MemRdB}_{cold\ start}^{(wave)}\right)_{num}\right)_{aux4} \times \text{L2MissRate}
\end{aligned}$$

$$\begin{aligned}
\left(\left(\text{MemRdB}_{cold\ start}^{(wave)}\right)_{num}\right)_{aux4} &= \text{L2RdB}_{total} \\
&- \left(\left(\left(\text{MemRdB}_{cold\ start}^{(wave)}\right)_{num}\right)_{aux4}\right)_{aux1}
\end{aligned}$$

$$\begin{aligned}
\left(\left(\left(\text{MemRdB}_{cold\ start}^{(wave)}\right)_{num}\right)_{aux4}\right)_{aux1} &= \text{Size}_B^{(matA)} \\
&+ \text{Size}_B^{(matA)} \times \left(\text{CEILING}\left(\frac{N_{dim}}{W_{(XeCUTile)}}, 1\right) - 1\right) \times P_{L2Miss, after\ 1st\ wave}^{(matA)} \\
&+ \text{Size}_B^{(matB)} \\
&+ \text{Size}_B^{(matB)} \times \left(\text{CEILING}\left(\frac{M_{dim}}{H_{(XeCUTile)}}, 1\right) - 1\right) \times P_{L2Miss, after\ 1st\ wave}^{(matB)}
\end{aligned}$$

row 117 - TOTAL HBM WRITE B

$$\text{MemWrB}^{(total)} = \text{Size}_B^{(matC, matD)}$$

row 118 - TOTAL HBM B

$$\text{MemRdWrB}^{(total)} = \text{MemWrB}^{(total)} + \text{MemRdB}_{cold\ start}^{(wave)}$$

row 121 - HBM READ B CLK

$$\text{MemRdBW}_{BpClk} = \frac{\text{MemRdB}_{cold\ start}^{(wave)}}{T_{clk}^{(total)}}$$

row 122 - HBM WRITE B CLK

$$\text{MemWrBW}_{BpClk} = \frac{\text{MemWrB}^{(total)}}{T_{clk}^{(total)}}$$

row 123 - HBM TOTAL B CLK

$$\text{MemRdWrBW}_{BpClk} = \frac{\text{MemRdWrB}^{(total)}}{T_{clk}^{(total)}}$$

row 126 - HBM BW PCT

$$\eta_{MemBW} = \frac{\text{MemRdWrBW}_{BpClk}}{\text{MemBW}_{BpClk}^{max}}$$

row 127 - GTI READ BW PCT

$$\eta_{GTIRdBW} = \frac{\text{MemRdBW}_{BpClk}}{\text{GtiRdBW}_{BpClk}^{max}}$$

row 128 - GTI WRITE BW PCT

$$\eta_{GTIW_rBW} = \frac{\text{MemWrBW}_{BpClk}}{\text{GtiWrBW}_{BpClk}^{max}}$$